Physiology of Normal Labor

Goal:

To empower participant with knowledge, skills, and a positive attitude toward labor process.

Objectives:

At the end of the lecture the student should be able to:

1) Recognize objectives of nursing care during 2nd stage of labor.

2) Give an example for woman's position during labor.

3) Summarize the preparations that needed for delivery.

4) Estimate when to transfer the woman to delivery room.

1) Perform the procedure effectively without assistance.

2) Create good and comprehensive nursing / patient relationship.

Introduction:

-Labor and delivery are the focus and the climax of the reproductive process. It's both a physical and psychological challenge for the mother and hazardous journey to the fetus. As well as bringing great joy and happiness to most of the family, labor has brought death and catastrophe to others so good care should be provided for both mother and fetus.

Definition of Labor:

Labor is the process by which the fetus, placenta and membrane are expelled through the birth canal.

Definition of Normal labor:

Spontaneous delivery of full-term, mature, living baby presenting by its vertex through the birth canal within a reasonable time without interference, without complication to the fetus and mother.

Theories of labor onset:

A) Hormonal theories:

- Estrogen theory.
- Progesterone withdrawal theory.
- Prostaglandin theory.
- Oxytocin theory.
- Fetal cortisol theory.

B) Mechanical theories:

- Uterine distention theory.
- Stretch of lower uterine segment.

1) Estrogen theory:

During pregnancy most of estrogen found in binding form but during last trimester freer estrogen appears, this increases the excitability of the uterus especially myometrium layer and this contractions lead to expulsion of uterine content.

2) Progesterone withdrawal theory:

Before labor, there is drop in progesterone synthesis leading to predominance of excitatory action of estrogen.

3) Prostaglandin theory:

Prostaglandin is a powerful stimulator for uterine muscles activity. Prostaglandin is released after rupture of amniotic membranes.

4) Oxytocin theory:

Pressure on the cervix stimulates the hypothesis to release oxytocin from the maternal posterior pituitary gland. As pregnancy advances, the uterus becomes more sensitive to oxytocin. The presence of this hormone causes the initiation of contraction of the smooth muscles of the body (the uterus is composed of smooth muscles).

5) Fetal cortisol theory:

Increased cortisol production from fetal adrenal gland before labor may influence its onset by increase production of estrogen from placenta.

Clinical pictures of labor:

- Prodromal (pre labor) stage:
  - Shelfing.
  - Lightening.
  - Increased pelvic pressure.
  - Increase vaginal discharge.

- False labor pain.
- The onset of labor:
  - ☞ True labor pain.
  - ☞ The show.
  - ☞ Cervical dilatation.
  - ☞ Formation of bag of fore water.
  - ☞ Desire of patient to evacuate the rectum.
  - ☞ Reflex desire to bear down.
  - ☞ Bearing down is accompanied by an expiratory grunt.
  - ☞ Rupture of membrane.

Prodromal (pre labor) stage:

- Shelfing:

  Is the falling forward of uterine funds making the upper abdomen looking like the shelf during standing position.

- Lightening:

  Is the relief of upper abdominal pressure symptoms like dyspnea, dyspepsia and palpitation due to:

  - Descend of fundal level after engagement of fetal head.
  - Shelfing of the uterus.
- Increased pelvic pressure:

  -When engagement of fetal head the women may experience the following symptoms frequency of urination, rectal tensums, difficulty of walking.

4) False labor pain:

- Also, can be described Braxton Hicks contractions as tightening in the abdomen that comes and goes. Many women say these “false” contractions feel like mild menstrual cramps. Braxton Hicks contractions may be uncomfortable, but they do not cause labor or open the cervix.

B- The onset of labor:

It is characterized by:

1. True labor pain
2. The bloody show: Vaginal drainage of brown, red blood tinged with mucous due to the release of the cervical mucus plug when the cervix dilated.
3. Dilatation of the cervix:

A closed cervix is a reliable sign that labor has not begun. In multigravida the cervix may admit the tip of the finger before onset of labor.

4. Formation of the bag of fore-waters:

It bulges through the cervix and becomes tense during uterine contractions.

-Factors Affecting the Progress of Labor:

1) Power:

- It's divided into:
  - Uterine contraction: the strength of uterine contraction along with the amniotic fluid between the fetal head and the maternal cervix during the descent plays a major

role in placing pressure on the cervical opening, contributing to uterine dilatation and assessment.

Differentiation between the true and false labour contractions

True contractions	False contractions
Begin irregularly, but become regular and predictable	Begin and remain irregular
Felt first in lower back and sweep around to the abdomen in a wave	Felt first abdominally and remain confined to the abdomen
Continue no matter what the women's level of activity	Often disappear with ambulation
Increase in duration, frequency, and intensity.	Do not increase in duration, frequency or intensity
Achieve cervical dilatation	Do not achieve cervical dilatation

- Mothers bearing down:

-Maternal ability to push and bear down is critical to the vaginal delivery of the newborn without external assistance.

2) **Passage:**

- The birth canal, including the pelvis, pelvic floor, cervix, vagina and vaginal opening.

2) **Passenger** (fetal as presentation, lie, attitude and position.

4) Psychological status of the mother:

Maternal anxiety and stress can impede the progress of labor by creating physiological changes that counteract the relaxation of pelvic structure as needed to promote fetal delivery.

- Positioning during labor and delivery:

  The maternal position can promote the use of gravity to assist in the fetal descent. This position can be:

Lithotomy position:

In the lithotomy position, the mother is lying on her back with her legs up in stirrups and her buttocks close to the edge of the table. This position is convenient for the caregiver because it permits him or her more access to the perineum. However, this is not a comfortable position for most patients, considering the pressure on the vaginal walls due to the fact that the baby's head is uneven, and the labor process is working against gravity.

Squatting position:

The squatting position increases pressure in the pelvic cavity with minimal muscular effort. The birth canal will open 20%-

30% more in a squat than in any other position. It is recommended for the second stage of childbirth.

All-fours:

Some mothers may choose the all-fours position instinctively. It can help the baby turn around in the case of a malpresentation of the head. Since this position uses gravity, it decreases back pain, as the mother is able to tilt her hips

Side lying:

Side lying may help slow the baby's descent down the birth canal, thereby giving the perineum more time to naturally stretch. To assume this position, the mother lies on her side with her knees bent. To push, a slight rolling movement is used such that the mother is propped up on one elbow is needed, while one leg is held up.

Stages of labor:

- **First stage: (stage of cervical dilatation)**
  - Start: with onset of true labor pains
  - End: with full dilatation of cervix
  - Duration: in prime gravida about 12 hours while In multigravida about 6:8 hours

Description of uterine contraction:

1- Frequency: - refers to number of contractions. It is the period between the beginnings of one contraction to the beginning of the next one.

2- **Interval**: - the period between the end of one contraction to the beginning of the next contraction.

3- **Duration**: It is measured from the beginning of increment to the completion of decrement.

4- **Intensity**: - Refers to strength of uterine contraction during acme (Mild, Mod, severe)

Signs (dilation of Cervix):

1) Dilatation: How far has the cervix opened (in cm). It ranged from zero to 10 measured by fingers as each finger equal 2 cm.

2) Effacement: How thin /soft and short is the cervix (in %).

First stage of labor is divided into:

1)- Latent phase:

- Cervix 0 cm to 3 cm dilatation.
- contraction are irregular and progress from mild to moderate, 5 to 30min apart, 30 to 45 seconds duration.

1. Partial cervical dilatation and effacement.
2. Mother is talkative and eager.

2) Active phase:

- Cervical dilatation from 4 cm to 10 cm.
- contractions are regular and progress from moderate to severe, 3 to 5min apart, 40 to 70 seconds duration.

- Fetal descend start and mother becomes more anxious and restlessness.
- Nausea and vomiting and sensation of needing to have a bowel movement may be noted.
- Urge to push noted.
- Bloody show increase as amniotic fluid escape.

Second stage of labor

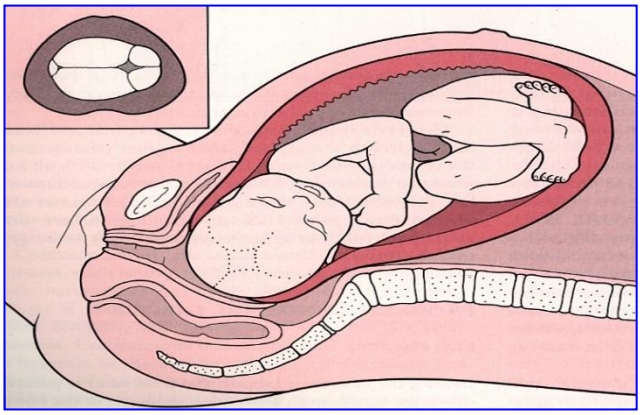
-It's the stage of fetal expulsion. It starts from full dilatation of the cervix to the delivery of the fetus.

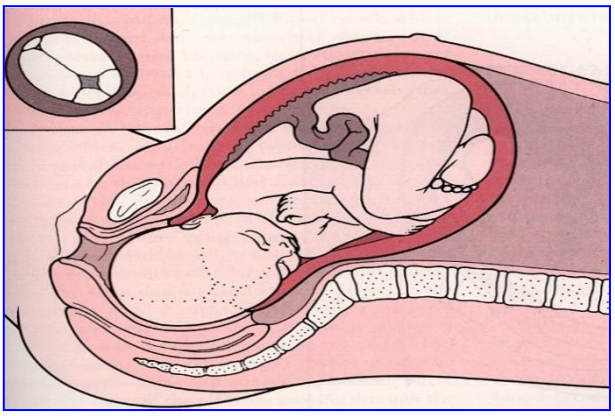
- Duration: in prime gravida up to 1 hour and in multigravida about 20 minutes.

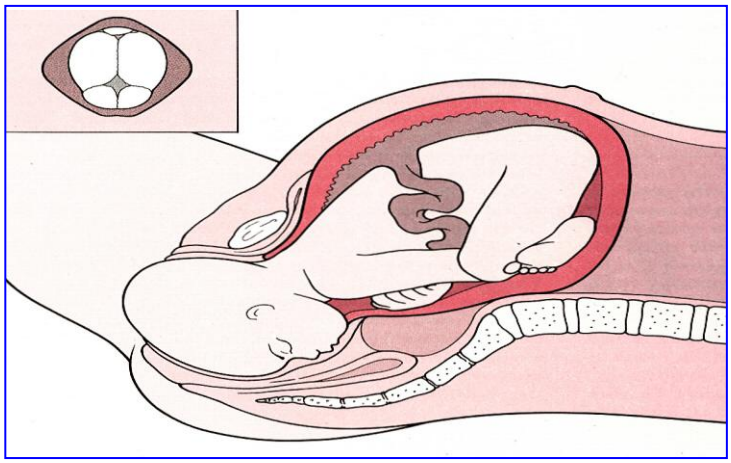
Signs of 2nd stage:

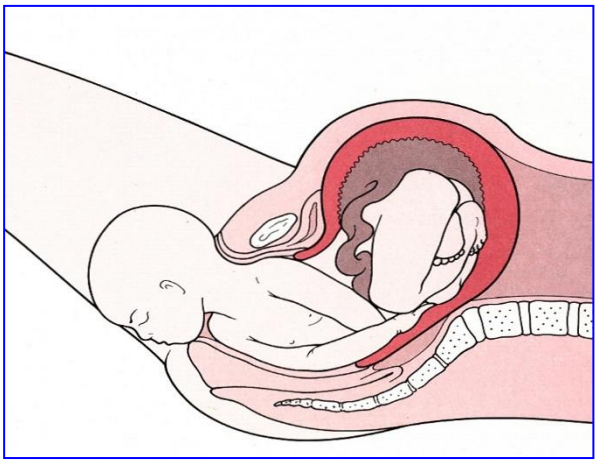
- Full dilatation of the cervix (4 inches or 5 fingers or 10 cm).
- Involuntary Bearing down during uterine contractions due to pressure of the presenting part on the rectum and pelvic floor causing reflex contractions of the axillary muscles.
- Bulging of the perineum and anal orifice start to dilate.
- Rupture of bag of water.
- Desire to defecate.
- Mother is irritable, unwilling to be touched and may cry.
- Grunting voice of the mother.

Mechanism of labor:









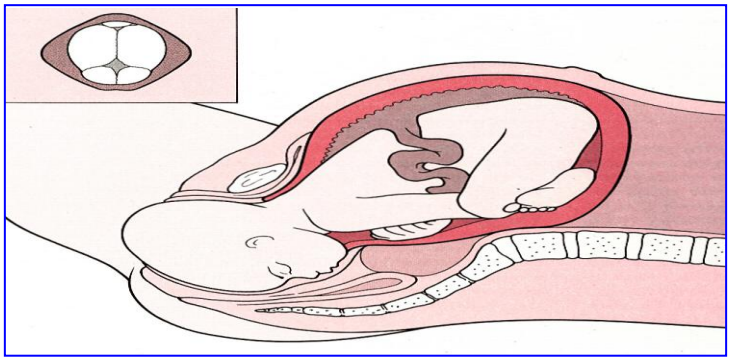


Figure: Mechanism of labor:

1) Descend:

-It is a constant movement during labor caused by:

- Uterine contractions and retractions
- Axillary forces of labor.

- **Engagement:**

The bi-parietal diameter of the fetal head had passed through the pelvic brim.

- **Increased flexion:**
  - As the head descends it meets resistance from the pelvic floor and the lateral pelvic walls causing increase flexion.
  - The head enters the pelvis with the smallest sub-occipital bregmatic.

Advantages:

- It brings the shortest longitudinal diameter of the head (Suboccipito-bregamtic = 9.5 cm) to pass through the birth canal.

- **Internal rotation:**
- After flexion has been completed, the occiput becomes the most dependent part.
- In occipito-anteior position the occiput rotates forwards 1/8 of a circle
- In occipital-posterior position the occiput rotates forwards 3/8 of a circle
- The occiput now hinges behind the symphysis pubis.

5) Extension:

- The supocciupto region lies under the symphysis pubis then by the head extension the vertex, forehead, and face coming extensively.

The head is affected by 2 forces:

- Uterine contraction that acting down and forward.
- The pelvic floor resistance acting up wards and forward so the net result is forward direction that result in extension of the head.

6) Restitution:

After delivery of the head, the occiput rotates 1/8 of a circle in the opposite direction to that of internal rotation to correct the twist of the neck.

7) External rotation:

1. The occiput undergoes another rotation 1/8 of a circle in the same direction to that of the restitution.
2. It is due to internal rotation of the anterior shoulder 1/8 of a circle which is transmitted to the delivered head.
3. So the final feature is that the occiput is directed to one thigh and the face towards the opposite thigh.

8) Delivery of the shoulder:

4. The anterior shoulder meets the pelvic floor first and rotates forwards 1/8 of a circle, with further descend, the anterior shoulder hinge below the symphysis pubis.

5. The posterior shoulder is delivered first by lateral flexion of the fetal spines then the anterior shoulder.
6. Finally, the trunk is delivered by lateral flexion of the fetal spines around the symphysis pubis.

Third stage of labour: **(**the stage of placental delivery**)**

It begins immediately after the baby is born, until the placenta is delivered, it takes about 20 minutes.

★ After delivery of the fetus, the uterine contractions and retractions continue in the same frequency as that of the second stage.

★ They are painless (Low amplitude), so the placental area diminishes from 20 cm to 7 cm, the placenta is inelastic and unable to shrink, so it starts to separate.

★ Expulsion of the placenta and membranes is helped by bearing down efforts of the mother.

Mechanism of placental delivery:

A. **Schultze mechanism**: (80 % of cases)

The central part of the placenta separates first because the placenta is firmly adherent at its edge.

Retro placental hematoma forms and the placenta is delivered by its fetal surface.

This mechanism is less liable for bleeding or retained fragments.

B. **Duncan mechanism**: (20 % of cases)

The placenta separates from its lower edge and delivers sideways it is more liable for bleeding or missed fragments.

Signs of placental separation:

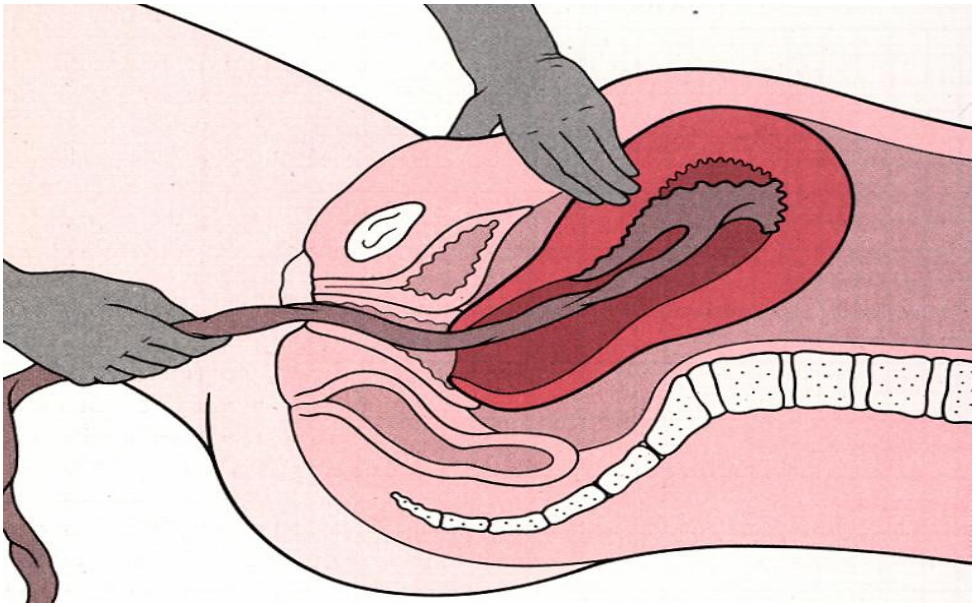
1-Gush of blood from vagina

2- Elongation of the cord

3- Loss of pulsation of the cord

4- Uterus becomes smaller, globular, and harder.

5- Supra-public bulge is noticed due to presence of placenta in the lower uterine segment.



placental separation

Fourth stage of labor: It is the recovery period after delivery of the fetus (2 hrs).